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Leading Causes of Death Among VHA Users

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Contributions

This fact sheet is a result of the efforts of the Boston Epidemiologic Research and Information Center.

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About CSPEAR

CSPEAR translates VA electronic health record data into brief, scientifically-reliable reports on the health status of Veterans. CSPEAR is a collaborative effort of the Cooperative Studies Program's national network of Epidemiology Centers.

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1. Introduction

Mortality data are used worldwide to document and track the health of populations, most notably by the World Health Organization (WHO) Global Burden of Disease (GBD) program¹ and in the U.S by the Centers for Disease Control (CDC), National Center for Health Statistics (NCHS). The Center for Disease Control (CDC) National Center for Health Statistics (NCHS) publishes a series of brief reports (MMWR, NVSR, NCHS data brief)^{2,3,4,5,6} annually revealing the leading causes overall and by age and gender rankings. This annual accounting of deaths allows the CDC to monitor trends, recognize emerging challenges, track the effectiveness of interventions, and make public health decisions that improve and save lives.

The objective of this work was to similarly determine a national accounting of the leading Causes of Death (COD) among Veteran Health Users (VHA) to inform priorities for health services research of the Veteran population, and to examine differences in cause by age and gender.

2 Methods

2.1 Population

We used the corporate data warehouse to identify Veteran Health Users in 2019 as follows:

Inclusion Criteria:

1. Confirmed as a Veteran based on the Veteran flag and eligibility for VA health care
2. Assigned to 1 of the 8 VA priority groups
3. Had consistent and complete numeric identifiers
4. Age ≥ 20 and ≤ 110 years; from DOB to January 1, 2019
5. Had at least 1 inpatient visit or outpatient visit (any kind) at a VHA facility recorded during 2018

Exclusion Criteria:

1. Patients with inconsistent Veteran statuses across or within their patient records
2. Test patients
3. Patients with a date of death prior to January 1, 2019

2.2 Death Data

Data on deaths occurring in 2019 for the Veteran Health Users identified from the CDW were procured from the Mortality Data Repository (MDR) which manages and distributes death data from the National Death Index (NDI) program within the Centers for Disease Control (CDC) National Center for Health Statistics (NCHS).

2.3 Classification of Phenotypes

The NCHS uses the cause of death reported on death certificates to determine the underlying cause and defined as “the disease or injury which initiated the train of morbid events leading directly to death, or the circumstances of the accident or violence which produced the fatal injury.”⁷

In practice, the underlying cause is selected from the conditions entered by the medical certifier on the death certificate, thus making the training and qualifications of medical certifiers critical for accurate documentation of causes of death⁸. When more than one cause or condition is entered, “the underlying cause is determined by the sequence of conditions on the certificate, provisions of ICD [International Classification of Diseases], and associated selection rules and modifications”⁷

Overall and sex specific mortality rates are presented in Table A. Causes of death were grouped and ranked according to CDC, NCHS developed phenotypes² used in their COD ranking of 113 selected causes of death. Results are presented overall and by age and sex in Tables B and C.

3 Results

3.1 Results: Tables

Table A: Veteran Health User Deaths in 2019: Overall and by Age and Birth Sex

Male				Female			
Deaths		VHA Users	Rate of Death per 100,000	Deaths		VHA Users	Rate of Death per 100,000
n	%			n	%		
267,956	100.0%	8,218,677	3,260	6,412	100.0%	828,804	774
67	0.03%	125,405	53	<11	--	37,201	--
1,070	0.40%	731,656	146	98	1.53%	159,302	62
1,651	0.62%	819,499	201	183	2.85%	186,449	98
4,150	1.55%	970,077	428	381	5.94%	164,953	231
18,898	7.05%	1,319,243	1,432	1,331	20.76%	173,592	767
67,466	25.18%	2,372,040	2,844	1,275	19.88%	73,765	1,728
67,799	25.30%	1,134,749	5,975	966	15.07%	19,931	4,847
106,855	39.88%	746,008	14,324	2,168	33.81%	13,611	15,928

Table B. 2019 Leading Causes of Death (based on ICD-10) for Male Veteran Health Users

Rank	Leading Cause of death *	Deaths		VHA Users	Rate of Death per 100,000
		n	%		
	Overall	267,956	100.0%	8,218,677	3,260
1	Heart diseases	71,169	26.6%	8,218,677	865.9
2	Cancers	57,071	21.3%	8,218,677	694.4
3	Chronic lower respiratory diseases	18,191	6.8%	8,218,677	221.3
4	CeVD	12,641	4.7%	8,218,677	153.8
5	Accidents	10,759	4.0%	8,218,677	130.9
6	Alzheimer disease	9,677	3.6%	8,218,677	117.7
7	Diabetes	9,127	3.4%	8,218,677	111.1
8	Nephritis, nephrotic syndrome and nephrosis	5,334	2.0%	8,218,677	64.9
9	Parkinson disease	5,166	1.9%	8,218,677	62.9
10	Influenza and pneumonia	4,758	1.8%	8,218,677	57.9
11	Septicemia	3,468	1.3%	8,218,677	42.2
12	Chronic liver disease and cirrhosis	3,263	1.2%	8,218,677	39.7
13	Suicide	3,092	1.2%	8,218,677	37.6
14	HTN and HTN renal disease	3,023	1.1%	8,218,677	36.8
15	Pneumonitis due to solids and liquids	2,430	0.9%	8,218,677	29.6
16	Benign tumors and tumors of unknown behavior	1,705	0.6%	8,218,677	20.7
17	Nutritional deficiencies	959	0.4%	8,218,677	11.7
18	Aortic aneurysm and dissection	890	0.3%	8,218,677	10.8
19	Complications of medical and surgical care	509	0.2%	8,218,677	6.2
20	Anemias	468	0.2%	8,218,677	5.7
21	Cholelithiasis and other disorders of gallbladder	437	0.2%	8,218,677	5.3
22	AthScl	422	0.2%	8,218,677	5.1
23	Assault (homicide)	393	0.1%	8,218,677	4.8
24	Enterocolitis due to Clostridium difficile	371	0.1%	8,218,677	4.5
25	Hep	338	0.1%	8,218,677	4.1
26	Peptic ulcer	302	0.1%	8,218,677	3.7
27	HIV	246	0.1%	8,218,677	3.0
28	Pneumoconioses and chemical effects	216	0.1%	8,218,677	2.6
29	Hernia	182	0.1%	8,218,677	2.2
30	Congenital malformations, deform & chrom abnormalities	156	0.1%	8,218,677	1.9
31	Hyperplasia of prostate	148	0.1%	8,218,677	1.8
32	Infections of kidney	56	0.0%	8,218,677	0.7
33	Diseases of appendix	37	0.0%	8,218,677	0.5
34	Meningitis	34	0.0%	8,218,677	0.4
35	TB	30	0.0%	8,218,677	0.4
36	Legal intervention	23	0.0%	8,218,677	0.3
37	Acute bronchitis and bronchiolitis	19	0.0%	8,218,677	0.2
38	Operations of war and their sequelae	14	0.0%	8,218,677	0.2

Table C. 2019 Leading Causes of Death (based on ICD-10) for Female Veteran Health Users

Rank	Leading Cause of death *	Deaths		VHA Users	Rate of Death per 100,000
		n	%		
	Overall	6,412	100.0%	828,804	774
1	Cancers	1,382	21.6%	828,804	166.7
2	Heart diseases	1,362	21.2%	828,804	164.3
3	Chronic lower respiratory diseases	457	7.1%	828,804	55.1
4	CeVD	345	5.4%	828,804	41.6
5	Alzheimer disease	335	5.2%	828,804	40.4
6	Accidents	294	4.6%	828,804	35.5
7	Diabetes	162	2.5%	828,804	19.5
8	Suicide	133	2.1%	828,804	16.0
9	Influenza and pneumonia	110	1.7%	828,804	13.3
10	Chronic liver disease and cirrhosis	100	1.6%	828,804	12.1
11	Nephritis, nephrotic syndrome and nephrosis	97	1.5%	828,804	11.7
12	HTN and HTN renal disease	91	1.4%	828,804	11.0
13	Septicemia	65	1.0%	828,804	7.8
14	Parkinson disease	54	0.8%	828,804	6.5
15	Nutritional deficiencies	36	0.6%	828,804	4.3
16	Pneumonitis due to solids and liquids	29	0.5%	828,804	3.5
17	Benign tumors and tumors of unknown behavior	26	0.4%	828,804	3.1
18	Assault (homicide)	24	0.4%	828,804	2.9
19	Complications of medical and surgical care	14	0.2%	828,804	1.7
20	Aortic aneurysm and dissection	13	0.2%	828,804	1.6
21	Cholelithiasis and other disorders of gallbladder	13	0.2%	828,804	1.6
22	AthScl	11	0.2%	828,804	1.3

4 References and Resources

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